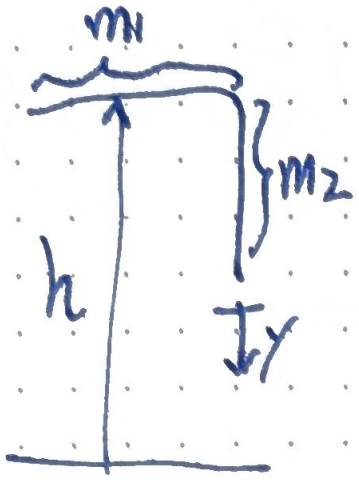


Midterm Q10:

$$T = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$$

$$= \frac{1}{2}m\dot{y}^2$$

$$U = m_1 g h_1 + m_2 g h_2$$

$$m_1 = m - \frac{m}{l}y$$

$$m_2 = \frac{m}{l}y$$

$$h_1 = h$$

$$h_2 = (h - \frac{y}{2})$$

$$U = (m - \frac{m}{l}y)gh + \frac{m}{l}y\,g(h - \frac{y}{2})$$

$$= mgh - \frac{mgh}{l}y + \frac{mgh}{l}y - \frac{m}{l}\frac{g}{2}y^2$$

$$= mgh - \frac{m\,g}{l\,2}y^2$$

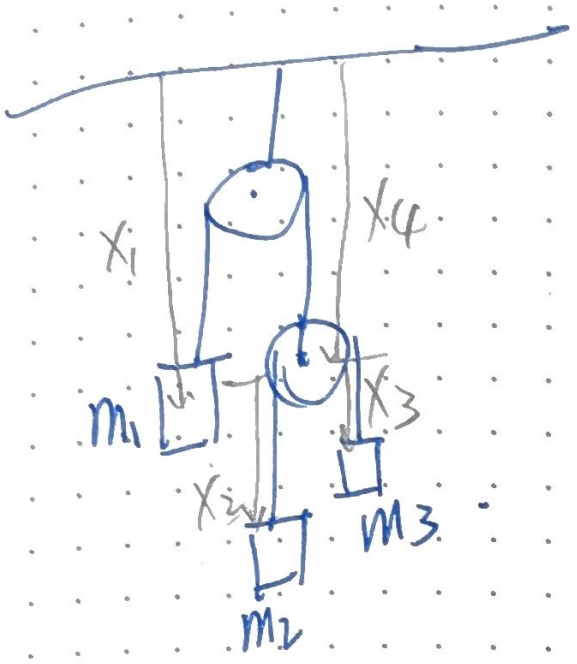
$$\mathcal{L} = T - V = \frac{1}{2}m\dot{y}^2 + \frac{m}{l}\frac{g}{2}y^2 - mgh$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y} \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\ddot{y}$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{m}{l}gy$$

E-L formula: $m\ddot{y} - \frac{m}{l}gy = 0$

Midterm Q9:

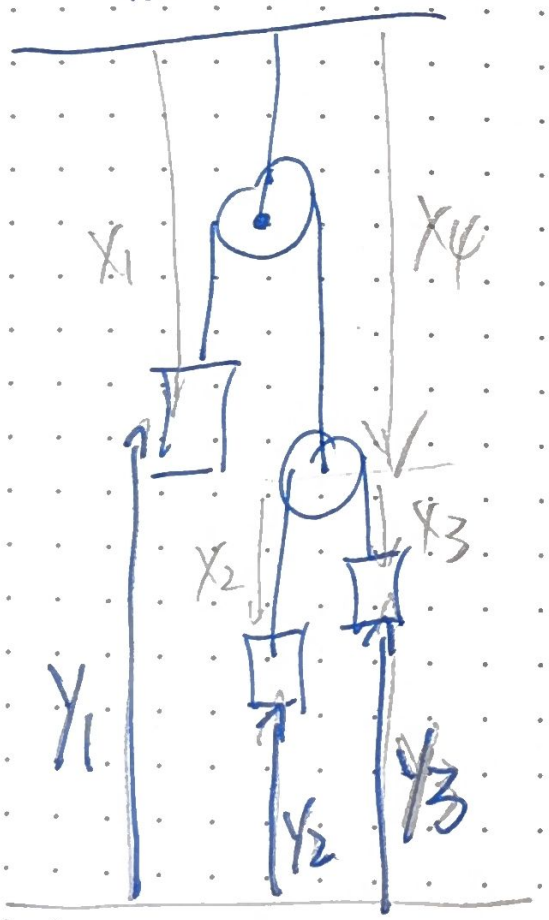


$x_1 + x_4 = l$

$x_2 + x_3 = l$

$\mathcal{L}(x_1, x_2)$ two DOF

BAD Pick of DOF, but still doable



extreme case



$x_1 + x_4 = l$

$y_1 + x_1 = 2l$

$y_2 + x_2 + x_4 = 2l$

$x_2 + x_3 = l$

$y_3 + x_3 + x_4 = 2l$

$\vdots$

$y_3 = 5l - 2y_1 + y_2$

$\mathcal{L}(y_1, y_2)$

$\rightarrow x_4 = l - x_1$
$= l - (2l - y_1)$
$= y_1 - l$

$x_2 = 2l - y_2 - x_4$
$= 2l - y_2 - y_1 + l$
$= 3l - y_2 - y_1$

$x_2 = 3l - y_1 - y_2$
$x_2 + x_3 = l$
$\rightarrow x_3 = l - x_2$
$= l - 3l + y_1 + y_2$
$= y_1 + y_2 - 2l$

$y_3 + x_3 + x_4 = 2l$
$\rightarrow y_3 = 2l - x_3 - x_4 = 2l - y_1 - y_2 + 2l - y_1 + l$
$= 5l - 2y_1 - y_2$

Lagrangian Mech.

Coordinates: q_i

Variable: q_i $\dot{q}_i$ $\left(\frac{d}{dt}q_i = \dot{q}_i\right)$

$$\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i, t)$$

"Action" $S = \int \mathcal{L}\, dt$

"Least action" ~~Stationary action~~"

$$\delta S = 0$$

Euler-Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0 \quad (Q_i)$$

Lagrangian Field Theory

q_i

$q_i, \dot{q}_i, q_i'$ $\frac{\partial}{\partial t}q_i = \dot{q}_i$ $\frac{\partial}{\partial x}q_i = q_i'$

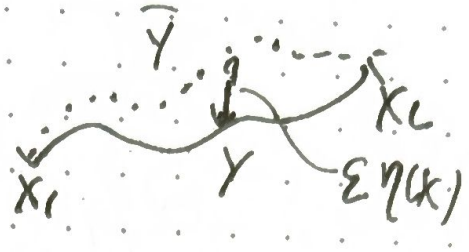
$\mathcal{L}(q_i, \dot{q}_i, q_i', t, x)$

"Action" $S = \int_{\text{time}}\int_{\text{space}} \mathcal{L}\, dt\, dx$

$\delta S = ?$

δ operator

Recall: $\bar{y} - y = \varepsilon\eta(x)$



$\delta y = \varepsilon\eta(x) = \bar{y} - y$

$\delta y' = \bar{y}' - y' = \varepsilon\eta'(x)$

The δ is commutative:

$\delta\left(\frac{\partial}{\partial x}\right) = \frac{\partial}{\partial x}(\delta y)$

$\delta \int y\,dx = \int \delta y\,dx$

$\delta(F(x,y,y')) = ?$

Taylor series expansion of $F(x,y,y')$ about δy

$$F(x, y+\delta y, y'+\delta y') = F(x,y,y') + \left(\frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'\right) + \emptyset$$

$$\underbrace{F(x, y+\delta y, y'+\delta y') - F(x,y,y')}_{\text{the def. of } \delta(F(x,y,y'))} = \frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y' \quad \delta^2 \text{ and above,}$$

we now have the multi-variational rule of the δ operation.

$$\delta S = \int_{t_1}^{t_2}\int_0^l \left(\frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i + \frac{\partial \mathcal{L}}{\partial q_i'}\delta q_i' \right) dx\,dt$$

$$= \int_{t_1}^{t_2}\int_0^l \left(\frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta\left(\frac{\partial q_i}{\partial t}\right) + \frac{\partial \mathcal{L}}{\partial q_i'}\delta\left(\frac{\partial q_i}{\partial x}\right) \right) dx\,dt$$

$$= \int_{t_1}^{t_2}\int_0^l \left(\frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\left(\frac{\partial \delta q_i}{\partial t}\right) + \frac{\partial \mathcal{L}}{\partial q_i'}\left(\frac{\partial \delta q_i}{\partial x}\right) \right) dx\,dt$$

$$= \underbrace{\int_{t_1}^{t_2}\int_0^l \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i\,dx\,dt}_{(1)} + \underbrace{\iint \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\left(\frac{\partial \delta q_i}{\partial t}\right) dx\,dt}_{(2)} + \underbrace{\iint \frac{\partial \mathcal{L}}{\partial q_i'}\left(\frac{\partial \delta q_i}{\partial x}\right) dx\,dt}_{(3)}$$

(1)

(2): $\iint \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{\partial(\delta q_i)}{\partial t}\,dt\,dx = \int_0^l \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta q_i \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial(\dot{q}_i)}\right)\delta q_i\,dt \right] dx$

(3) $\iint \frac{\partial \mathcal{L}}{\partial q_i'}\frac{\partial(\delta q_i)}{\partial x}\,dx\,dt = \int_{t_1}^{t_2} \left[\frac{\partial \mathcal{L}}{\partial q_i'}\delta q_i \Big|_0^l - \int_0^l \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial(q_i')}\right)\delta q_i\,dx \right] dt$

chain rule and integration by parts in both $\frac{\partial}{\partial t}$ and $\frac{\partial}{\partial x}$

$$\delta(S) = \iint \left(\frac{\partial \mathcal{L}}{\partial q_i} \delta q_i \, dx dt - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) \delta q_i \, dt dx - \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial q_i'}\right) \delta q_i \, dx dt \right)$$

$$= \iint \left(\frac{\partial \mathcal{L}}{\partial q_i} - \frac{\partial}{\partial t}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial q_i'} \right) \delta q_i \, dx dt = 0$$

$\therefore \delta q_i$ is arbitrary

$$\therefore \frac{\partial \mathcal{L}}{\partial q_i} - \frac{\partial}{\partial t}\frac{\partial \mathcal{L}}{\partial \left(\frac{\partial q_i}{\partial t}\right)} - \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial \left(\frac{\partial q_i}{\partial x}\right)} = 0$$

Hamilton's principle in 2 field theory form

or in the form of:

$$\frac{\partial}{\partial t}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} + \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial q_i'} - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$$\dot{q}_i = \frac{\partial q_i}{\partial t}$$

$$q_i' = \frac{\partial q_i}{\partial x}$$

or in the form of

$$\frac{\partial}{\partial t}\frac{\partial \mathcal{L}}{\partial (\partial_t q_i)} + \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial (\partial_x q_i)} - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$